

Exploring RGB and Video V-JEPA 2.1 Features for Aerial Depth Estimation and Semantic Scene Completion

Table 6. Additional depth controls extending main-paper Table 3.

Method	$\delta_1 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	SILog $\downarrow$
VJ DPT	0.747	0.162	6.730	0.200
VJ video Linear Probe	0.701	0.191	7.055	0.207
D3 Linear Probe	0.782	0.156	6.097	0.186
D3 FiLM-DPT	0.848	0.121	4.358	0.161
VJ DTA-MVS	0.654	0.231	8.197	24.717
VJ DTA-FiLM-MVS	0.768	0.175	6.406	0.211
MVSF++/D2-B	0.748	0.231	7.657	0.195
MVSF++/VJ-G	0.761	0.225	7.517	0.193
MVSF++/D3-H+	0.828	0.136	4.611	0.130

Table 7. Altitude interpolation diagnostics.

Method	$\delta_1 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	SILog $\downarrow$
Tr: 01–08@30/50 m; Val: 01–04@40 m; Te: 05–08@40 m					
DPT	0.789	0.151	6.728	5.775	0.180
FiLM-DPT	0.965	0.102	4.707	3.957	0.120

Table 8. FiLM conditioning ablation. Test mean $\pm$ SD over three seeds; best mean in bold.

Conditioning	$\delta_1 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	SILog $\downarrow$
DPT	.728 $\pm$.034	.174 $\pm$.006	6.709 $\pm$.385	.202 $\pm$.010
Altitude FiLM	.798 $\pm$.020	.145 $\pm$.005	5.236 $\pm$.322	.180 $\pm$.006
Intrinsics FiLM	.711 $\pm$.015	.182 $\pm$.004	7.115 $\pm$.243	.216 $\pm$.007
Alt.+intr. FiLM	.810$\pm$.006	.137$\pm$.005	4.907$\pm$.140	.171$\pm$.005

LD5V: Spatial alignment: Frozen RAFT estimates target-to-source flow and warps neighboring 2D features into target-frame coordinates. For a stronger MVS comparison, MVSF++ uses cross-view fusion, attention-based encoding, and cost-volume regularization. Extending main-paper Table 5, its D3 and VJ variants integrated into FSSC yield SC IoU/SSC mIoU of 43.37/4.47% and 35.02/3.28%, respectively. Replacing estimated depth with GT metric depth only during geometric lifting yields 49.40/7.25% with VJ context and 42.01/8.40% with D3 context. Per-class results are provided in Supplement B; MVSF++-FSSC variants will be added.

CWVJ: Suggested D3 comparison also motivates, if permitted, a revised title: “Exploring Foundation-Model Spatial and Temporal Features for Aerial Depth Estimation and Semantic Scene Completion.” Table 8 for FiLM contribution ablations given the limited rebuttal period, and can extend this analysis to all published results. Our study focuses on frozen-feature transfer for MDE and SSC rather than onboard deployment; we therefore defer SWaP profiling to compact UAV-suitable student models and will add the suggested related work. Existing aerial datasets with metric depth can extend MDE evaluation, but those without voxel-level annotations cannot support SSC evaluation, for which OccuFly provides the required ground-truth voxel grids.

y7qY: Low SSC mIoU persists with GT depth (7.25/8.40), indicating bottlenecks beyond depth; learning is further challenged by OccuFly’s 96.109% empty versus 3.891% occupied valid voxels. The video result shows no benefit from our current adapter—not that temporal cues are ineffective—motivating motion/pose-aligned, occlusion-aware fusion. MVSFormer++ provides stronger cross-view geometry than our simple MVS module. The altitude-interpolation ablation uses scenes 1–8 only.

We thank all reviewers for their carefully reviewed constructive feedback. VJ/D2/D3/DA2/MVSF abbreviate V-JEPA 2.1/DINOV2/DINOV3/Depth Anything 2/MVSFormer++; response table numbering starts at 6, continuing submission.

Responses common to LD5V, CWVJ, and y7qY: Table 6 adds the corrected DPT result, spatially aligned video methods, DINOv3 comparison, simple-MVS, and supportive sophisticated MVSF++ variants. For space, we omit MAE, δ_2 , δ_3 , and published baselines already reported in OccuFly Table 5, retaining DA2-OccuFly as the key comparator. We withdraw the VJ FiLM-DPT superiority claim: Although DAv2-OccuFly achieves the strongest reported SiLog and threshold accuracies, foundation models with general pretraining objectives remain competitive and outperform it on absolute-error metrics. Any remaining gap may reflect DAv2’s depth-specific pretraining and end-to-end OccuFly adaptation, whereas frozen VJ/D backbones are adapted only through their depth heads.

Table 7 shall be Table 4. This non-standard split isolates interpolation to an unseen altitude while retaining familiar scenes; Suggested leakage is real and hence a new ablation with unseen scene and altitude will be added. We will replace Figures 1 and 5 with a clear end-to-end SSC data-flow diagram.